

AI Face Recognition - Front Cam Login/Logout System Log

1. Architecture: Single Front Camera (Index 0) automated authentication pipeline
2. Core Stack: PyTorch, facenet-pytorch (MTCNN + ResNet), OpenCV, NumPy, Pillow
3. First Arrival: User detected for the first time triggers automated LOGIN
4. Cooldown Mechanism: 10-second safety buffer prevents false immediate triggers
5. Re-entry Action: After cooldown, returning user automatically triggers LOGOUT
6. Dynamic Purging: Previous image cropped file auto-purged, updated with latest frame
7. Activity Tracking: Every Login and Logout is appended to attendance_log.csv
8. Visual Feedback: Green box on screen for LOGIN, Red box overlay for LOGOUT